Sri Lanka Institute of Information Technology



SE2030 – Software Engineering
Year 2, Semester 1- 2026

<Web based School Information System>

Agile Development – Lab 02

Group <2026-Y2-S1-MLB-B1G2-09>

IT Number	Name
IT 25 1020 25	Abishek S.A.
IT 25 1029 57	Godigamuwa G.J.E.
IT 25 1041 13	Sanjai G.
IT 25 1021 49	Dissanayake W.M.P.G.S.D.
IT 25 2001 67	Wijewardana K.M.N.J.
IT 25 1011 44	Adikari A.A.C.J.M.

1.User Stories

1.Principal

- As a Principal, I want to view academic performance reports across all classes, so that I can identify areas needing improvement.
- As a Principal, I want to view school attendance reports, so that I can monitor student and teacher attendance.
- As a Principal, I want to view the school's overall financial summary, so that I can keep track of fee collections and pending dues.
- As a Principal, I want to assign and manage support tickets so that reported issues are handled by the appropriate staff.
- As a Principal, I want to view class timetables, so that I can monitor the school's academic schedule.

2. Teacher

- As a Teacher, I want to mark daily attendance for my class, so that attendance records stay accurate and up to date.
- As a Teacher, I want to enter and update examination marks so that student academic performance can be maintained accurately.
- As a Teacher, I want to view my assigned classes and subjects, so that I know my teaching schedule and responsibilities.
- As a Teacher, I want to raise a ticket for an operational issue, so that it gets resolved by the right staff member.
- As a Teacher, I want to view student performance, so that I can identify students who need improvement.
- As a Teacher, I want to view attendance history for my classes, so that I can monitor student attendance over time.

3.Finance Staff

- As a Finance Staff, I want to set up fee structures for each class and term, so that fee amounts stay consistent and there's no confusion later.
- As a Finance Staff, I want to record fee payments made by students, so that all payments are accurately tracked in the system.
- As a Finance Staff, I want to view outstanding and overdue payments, so that I can follow up with students who have unpaid fees.
- As a Finance Staff, I want to generate a financial summary report, so that the school management can get a clear picture of how much fees have been collected.
- As a Finance Staff, I want to search and filter fee and payment records, so that I can quickly locate specific financial information.
- As a Finance Staff, I want to correct a previously recorded payment, so that data entry errors can be fixed without losing accountability.

4. Administrator

- As an Administrator, I want to create, edit, and deactivate staff accounts, so that I can control who has access to the system.
- As an Administrator, I want to assign roles and permissions to staff accounts, so that each user can only access the features relevant to their responsibilities.
- As an Administrator, I want to set up classes and sections and assign a class teacher to each one, so the academic structure of the school is properly organized.
- As an Administrator, I want to create timetables with automatic conflict checking, so that no teacher or classroom ends up double-booked by mistake.
- As an Administrator, I want to add subjects and assign them to the right classes and teachers, so the curriculum setup matches what's actually being taught.
- As an Administrator, I want to create exam schedules and allocate exam halls to each class/subject, so that exams are conducted without timing conflicts or overcrowded venues.
- As an Administrator, I want the system to automatically send notification alerts for things like overdue fees or upcoming exams, so staff don't have to keep checking manually.

2. Product Backlog

ID	User Story	Priority
PBI - 01	As a Principal, I want to view academic performance reports across all classes, so that I can identify areas needing improvement.	High
PBI - 02	As a Principal, I want to view school attendance reports, so that I can monitor student and teacher attendance.	High
PBI - 03	As a Principal, I want to view the school's overall financial summary, so that I can keep track of fee collections and pending dues.	High
PBI - 04	As a Teacher, I want to mark daily attendance for my class, so that attendance records stay accurate and up to date.	High
PBI - 05	As a Teacher, I want to enter and update examination marks so that student academic performance can be maintained accurately.	High
PBI - 06	As a Teacher, I want to view my assigned classes and subjects, so that I know my teaching schedule and responsibilities.	High

PBI - 07	As a Finance Staff, I want to set up fee structures for each class and term, so that fee amounts stay consistent and there's no confusion later.	High
PBI - 08	As a Finance staff, I want to record fee payments made by students, so that all payments are accurately tracked in the system.	High
PBI - 09	As a Finance staff, I want to view outstanding and overdue payments, so that I can follow up with students who have unpaid fees.	High
PBI - 10	As an Administrator, I want to create, edit, and deactivate staff accounts, so that I can control who has access to the system.	High
PBI - 11	As an Administrator, I want to assign roles and permissions to staff accounts, so that each user can only access the features relevant to their responsibilities	High
PBI – 12	As an Administrator, I want to set up classes and sections and assign a class teacher to each one, so the academic structure of the school is properly organized.	High
PBI – 13	As an Administrator, I want to create timetables with automatic conflict checking, so that no teacher or classroom ends up double-booked by mistake.	High
PBI – 14	As an Administrator, I want to add subjects and assign them to the right classes and teachers, so the curriculum setup matches what's actually being taught.	High
PBI – 15	As a Principal, I want to assign and manage support tickets so that reported issues are handled by the appropriate staff.	Medium
PBI – 16	As a Teacher, I want to raise a ticket for an operational issue, so that it gets resolved by the right staff member.	Medium
PBI – 17	As a Teacher, I want to view student performance, so that I can identify students who need improvement.	Medium
PBI – 18	As a Finance Staff, I want to generate a financial summary report, so that the school management can get a clear picture of how much fees have been collected.	Medium

PBI – 19	As an Administrator, I want to create exam schedules and allocate exam halls to each class/subject, so that exams are conducted without timing conflicts or overcrowded venues.	Medium
PBI – 20	As a Principal, I want to view class timetables, so that I can monitor the school's academic schedule.	Low
PBI – 21	As a Teacher, I want to view attendance history for my classes, so that I can monitor student attendance over time.	Low
PBI – 22	As a Finance Staff, I want to search and filter fee and payment records, so that I can quickly locate specific financial information.	Low
PBI – 23	As a Finance Staff, I want to correct a previously recorded payment, so that data entry errors can be fixed without losing accountability.	Low
PBI – 24	As an Administrator, I want the system to automatically send notification alerts for things like overdue fees or upcoming exams, so staff don't have to keep checking manually.	Low

3.Sprint Plan

1st Sprint – Week 4 to 7

Sprint ID	User Story	Priority
SP1 - 01	As a Principal, I want to view academic performance reports across all classes, so that I can identify areas needing improvement.	High
SP1 - 02	As a Principal, I want to view school attendance reports, so that I can monitor student and teacher attendance.	High
SP1 - 03	As a Principal, I want to view the school's overall financial summary, so that I can keep track of fee collections and pending dues.	High
SP1 - 04	As a Teacher, I want to mark daily attendance for my class, so that attendance records stay accurate and up to date.	High
SP1 - 05	As a Teacher, I want to enter and update examination marks so that student academic performance can be maintained accurately.	High
SP1 - 06	As a Teacher, I want to view my assigned classes and subjects, so that I know my teaching schedule and responsibilities.	High

Sprint Backlog – 1st Sprint

1. View Academic Performance reports

Task ID	Task	Est. Hours
T - 01.1	Design academic performance report view (by class/subject)	3
T - 01.2	Develop API to aggregate marks and grade data	2
T - 01.3	Add trend visualization (term-over-term comparison)	5
T - 01.4	Test academic report accuracy	2

2. View Attendance Reports

Task ID	Task	Est. Hours
T - 02.1	Design attendance report dashboard	4
T - 02.2	Develop API to retrieve attendance data	6
T - 02.3	Add filters by class and date	3
T - 02.4	Display student and teacher attendance summaries	3
T - 02.5	Test attendance report functionality	3

3. View Financial Summary

Task ID	Task	Est. Hours
T - 03.1	Design Principal-level financial summary view	3
T - 03.2	Configure role-based access to the financial report API	3
T - 03.3	Display collected fees versus outstanding dues	2
T - 03.4	Test Principal financial view access and accuracy	2

4. Mark Daily Attendance

Task ID	Task	Est. Hours
T - 04.1	Design daily attendance marking UI (per class/section)	4
T - 04.2	Develop backend API to save attendance records	3
T - 04.3	Implement present/absent/late status options	2
T - 04.4	Test attendance marking flow	2

5. Examination Marks

Task ID	Task	Est. Hours
T - 05.1	Design examination and marks management interface	4
T - 05.2	Implement marks entry and updating	3
T - 05.3	Implement marks validation and grade calculation	2
T - 05.4	Develop student performance viewing	2
T - 05.5	Integrate and test examination workflow	2

6. View Assigned Classes & Subjects

Task ID	Task	Est. Hours
T - 06.1	Design "My Classes" view for teacher dashboard	4
T - 06.2	Develop API to fetch teacher's assigned classes/subjects	3
T - 06.3	Display class list with student counts	2
T - 06.4	Test assigned classes view accuracy	2

2nd Sprint – Week 8 to 10

Sprint ID	User Story	Priority
SP2 - 01	As a Finance Staff, I want to set up fee structures for each class and term, so that fee amounts stay consistent and there's no confusion later.	High
SP2 - 02	As a Finance staff, I want to record fee payments made by students, so that all payments are accurately tracked in the system.	High
SP2 - 03	As a Finance staff, I want to view outstanding and overdue payments, so that I can follow up with students who have unpaid fees.	High
SP2 - 04	As an Administrator, I want to create, edit, and deactivate staff accounts, so that I can control who has access to the system.	High
SP2 - 05	As an Administrator, I want to assign roles and permissions to staff accounts, so that each user can only access the features relevant to their responsibilities.	High
SP2 - 06	As an Administrator, I want to set up classes and sections and assign a class teacher to each one, so the academic structure of the school is properly organized.	High

Sprint Backlog – 2nd Sprint

1. Define fee Structures

Task ID	Task	Est. Hours
T - 01.1	Design fee structure creation form (class, term, fee type, amount)	3
T - 01.2	Develop backend API to create, update, and retrieve fee structures	3
T - 01.3	Add validation to prevent duplicate or negative fee entries	2
T - 01.4	Store fee structure records in the database	2
T - 01.5	Test fee structure creation and update flow	2

2. Record fee payments

Task ID	Task	Est. Hours
T - 02.1	Design payment entry form (student, amount, date, payment method)	4
T - 02.2	Develop backend API to save payment records	4
T - 02.3	Link payment record to the student's financial profile	3
T - 02.4	Auto-update outstanding balance after payment	2
T - 02.5	Test payment recording functionality	2

3. Track outstanding & Overdue Payments

Task ID	Task	Est. Hours
T - 03.1	Design outstanding/overdue payments dashboard	4
T - 03.2	Develop API to calculate and fetch outstanding balances	4
T - 03.3	Implement flagging logic for accounts overdue beyond set days	3
T - 03.4	Add filters by class, term, and overdue status	2
T - 03.5	Test outstanding payment tracking	2

4. Manage Staff Accounts

Task ID	Task	Est. Hours
T - 04.1	Design staff account creation/edit UI	4
T - 04.2	Develop backend API for account CRUD operations	3
T - 04.3	Implement account status management (active/inactive)	3
T - 04.4	Add validation to prevent duplicate accounts (e.g., duplicate email)	2
T - 04.5	Test account creation, edit, and deactivation flow	2

5. Assign roles & permissions

Task ID	Task	Est. Hours
T - 05.1	Define role-permission matrix for all four staff roles	4
T - 05.2	Implement role-based access control (RBAC) in the backend	4
T - 05.3	Build UI for assigning and updating a user's role	3
T - 05.4	Restrict module access based on assigned role	2
T - 05.5	Test access boundaries for each role	2

6. Class & Section Management

Task ID	Task	Est. Hours
T - 06.1	Design class/section creation UI	4
T - 06.2	Develop backend API to create classes and sections	3
T - 06.3	Implement class teacher assignment feature	3
T - 06.4	Add class capacity setting	2
T - 06.5	Test class/section setup flow	2

3rd Sprint – Week 11 to 12

Sprint ID	User Story	Priority
SP3 - 01	As an Administrator, I want to create timetables with automatic conflict checking, so that no teacher or classroom ends up double-booked by mistake.	High
SP3 - 02	As an Administrator, I want to add subjects and assign them to the right classes and teachers, so the curriculum setup matches what's actually being taught.	High
SP3 - 03	As a Principal, I want to assign and manage support tickets so that reported issues are handled by the appropriate staff.	Medium
SP3 - 04	As a Teacher, I want to raise a ticket for an operational issue, so that it gets resolved by the right staff member.	Medium
SP3 - 05	As a Teacher, I want to view student performance, so that I can identify students who need improvement.	Medium
SP3 - 06	As a Finance Staff, I want to generate a financial summary report, so that the school management can get a clear picture of how much fees have been collected.	Medium

Sprint Backlog – 3rd Sprint

1. Build conflict-free timetables

Task ID	Task	Est. Hours
T - 01.1	Design timetable creation UI (weekly grid)	4
T - 01.2	Develop backend API to save timetable entries	3
T - 01.3	Implement conflict-checking logic (teacher/room double-booking)	2
T - 01.4	Add room allocation feature	2
T - 01.5	Test timetable creation and conflict detection	2

2. Subject Management

Task ID	Task	Est. Hours
T - 02.1	Design subject creation/assignment UI	3
T - 02.2	Develop backend API to define subjects	3
T - 02.3	Implement subject-to-class assignment	2
T - 02.4	Implement subject-to-teacher assignment	2
T - 02.5	Test subject assignment functionality	2

3. Ticket assignment & management

Task ID	Task	Est. Hours
T - 03.1	Design ticket management dashboard	3
T - 03.2	Implement ticket assignment to relevant staff	3
T - 03.3	Implement ticket priority, status and escalation management	2
T - 03.4	Implement ticket response, resolution and closure	2
T - 03.5	Integrate notifications and test ticket management workflow	2

4. Raise operational Ticket

Task ID	Task	Est. Hours
T - 04.1	Design ticket raising form (category, description, priority)	3
T - 04.2	Develop API to create and submit tickets	3
T - 04.3	Link ticket to the raising teacher's profile	2
T - 04.4	Test ticket submission flow	2

5. View Student Performance

Task ID	Task	Est. Hours
T - 04.1	Design student performance view	3
T - 04.2	Develop API to retrieve student results	3
T - 04.3	Display marks and grades	2
T - 04.4	Add performance filtering by student/subject	2
T - 04.5	Test student performance view	2

6. Generate financial report

Task ID	Task	Est. Hours
T - 04.1	Design financial summary report layout	3
T - 04.2	Develop API to aggregate fee collection data	5
T - 04.3	Implement report filters (by term, by class)	2
T - 04.4	Add export option (PDF/print)	2
T - 04.5	Test report generation accuracy	2

4th Sprint – Week 13 to 14

Sprint ID	User Story	Priority
SP4 - 01	As an Administrator, I want to create exam schedules and allocate exam halls to each class/subject, so that exams are conducted without timing conflicts or overcrowded venues.	Medium
SP4 - 02	As a Principal, I want to view class timetables, so that I can monitor the school's academic schedule.	Low
SP4 - 03	As a Teacher, I want to view attendance history for my classes, so that I can monitor student attendance over time.	Low
SP4 - 04	As a Finance Staff, I want to search and filter fee and payment records, so that I can quickly locate specific financial information.	Low
SP4 - 05	As a Finance Staff, I want to correct a previously recorded payment, so that data entry errors can be fixed without losing accountability.	Low
SP4 - 06	As an Administrator, I want the system to automatically send notification alerts for things like overdue fees or upcoming exams, so staff don't have to keep checking manually.	Low

Sprint Backlog – 4th Sprint

1. Exam Schedule + Hall allocation without conflict

Task ID	Task	Est. Hours
T - 01.1	Design and develop exam schedule UI (date, time, class, subject, hall selection)	6
T - 01.2	Develop backend API for exam schedule and hall management (CRUD)	3
T - 01.3	Implement conflict-checking logic (timing conflicts + hall double-booking)	3
T - 01.4	Implement hall capacity validation and allocation logic	2
T - 01.5	Test exam scheduling and hall allocation flow (including conflict scenarios)	2

2. View class timetables

Task ID	Task	Est. Hours
T - 02.1	Design timetable viewing interface	6
T - 02.2	Develop API to retrieve class timetables	3
T - 02.3	Add class/section filters	3
T - 02.4	Display timetable details	2
T - 02.5	Test timetable viewing functionality	2

3. View attendance history

Task ID	Task	Est. Hours
T - 03.1	Design attendance history interface	4
T - 03.2	Develop API to retrieve attendance history	3
T - 03.3	Add filters by student and date	2
T - 03.4	Display attendance history	2
T - 03.5	Test attendance history functionality	2

4. Search and filter financial records

Task ID	Task	Est. Hours
T - 04.1	Design search/filter interface (by student name, ID, class, date)	4
T - 04.2	Develop backend search API with multiple filter parameters	3
T - 04.3	Display search results in a structured, sortable table	2
T - 04.4	Test search accuracy and speed	2

5. Correct payment records

Task ID	Task	Est. Hours
T - 05.1	Design search/filter interface (by student name, ID, class, date)	4
T - 05.2	Develop backend search API with multiple filter parameters	3
T - 05.3	Display search results in a structured, sortable table	2
T - 05.4	Test search accuracy and speed	2

6. Automated notifications

Task ID	Task	Est. Hours
T - 06.1	Design notification trigger settings (event types)	4
T - 06.2	Integrate email service for sending notifications	3
T - 06.3	Implement triggers (overdue fees, exam schedules)	2
T - 06.4	Test email notifications for each event type	2